

Binomial Distribution

It is concerned with trials of a repetitive nature in which only the occurrence or non-occurrence, success or failure, acceptance or rejection, yes or no of a particular event is of interest.

- If we perform a series of independent trials such that for each trial, P

Probability of success = P

Probability of failure = $q = 1 - P$

Then the probability of r success in a series of n trials is given

by ${}^nC_r P^r q^{n-r}$, where r takes any integral value of from 0 to n

- Mean = np

- Variance = $npq = \mu_2 = \sigma^2$

- $\mu_3 = npq(q-p)$

- $\mu_4 = npq[1 + 3(n-2)pq]$

- $\beta_1 = \frac{\mu_3}{\mu_2^3}$

- $\beta_2 = \frac{\mu_4}{\mu_2^2}$

- Variance < Mean

• If n independent trials constitute one experiment & this experiment is repeated N times then the frequency of r successes is $N {}^n C_r P^r q^{n-r}$.

Q:- The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that

- exactly two will be defective
- at least two will be defective
- none will be defective.

Solⁿ:- Here $n = 12$
 Probability of a defective pen
 $= P = \frac{1}{10} = 0.1$

Probability of a non-defective pen $= q = 1 - \frac{1}{10} = \frac{9}{10} = 0.9$

(a) Probability that exactly two will be defective
 $P(X=2) = {}^{12}C_2 (0.1)^2 (0.9)^{10} = 0.2301$

(b) Probability that atleast two will be defective $= P(X \geq 2) = 1 - P(X \leq 1)$
 $= 1 - [\text{probability that either none or one is non-defective}]$

$$= 1 - \left[{}^{12}C_0 (0.9)^{12} + {}^{12}C_1 (0.1)(0.9)^{11} \right]$$

$$= 0.3412$$

(c) Probability that none will be defective $= {}^{12}C_0 (0.9)^{12} = 0.2833$
 $= P(X=0)$

Q:- In 256 sets of 12 tosses of a coin, in how many cases one can expect 8 heads & 4 tails.

Solⁿ:- Here $n = 12$, $N = 256$

$p =$ prob. of head $= \frac{1}{2}$

$q =$ prob. of tail $= \frac{1}{2}$

Prob. of 8 heads & 4 tails

$$= P(X=8) = {}^{12}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^4 = \frac{495}{4096}$$

The expected no. of such cases

$$= N \cdot P(X=8) = 256 \times \frac{495}{4096} = 30.9$$

$$\approx 31$$

Hence, in this case, the binomial distribution is unimodal, the unique mode being np .

If np is a whole number (i.e., integer), then the distribution is unimodal and the mean and mode are equal, each being np .

Example 14.1. Ten unbiased coins are tossed simultaneously. Find the probability of obtaining,

- (i) Exactly 6 heads (ii) At least 8 heads (iii) No head
(iv) At least one head (v) Not more than three heads (vi) At least 4 heads

Solution. If p denotes the probability of a head, the $p = q = \frac{1}{2}$. Here $n = 10$. If the random variable X denotes the number of heads, then by the Binomial probability law, the probability of r heads is given by,

$$\begin{aligned} p(r) &= P(X = r) = {}^nC_r p^r \cdot q^{n-r} \\ &= {}^{10}C_r \left(\frac{1}{2}\right)^r \cdot \left(\frac{1}{2}\right)^{10-r} = {}^{10}C_r \cdot \left(\frac{1}{2}\right)^{10} = \frac{1}{1024} {}^{10}C_r \end{aligned} \quad \dots (*)$$

- (i) Required probability $= p(6) = \frac{1}{1024} {}^{10}C_6 = \frac{210}{1024} = \frac{105}{512}$ [From (*)]
 (ii) Required probability $= P(X \geq 8) = p(8) + p(9) + p(10)$
 $= \frac{1}{1024} [{}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}]$ [From (*)]
 $= \frac{45 + 10 + 1}{1024} = \frac{56}{1024} = \frac{7}{128}$
 (iii) Required probability $= P(X = 0) = p(0) = \frac{1}{1024} {}^{10}C_0 = \frac{1}{1024}$ [From (*)]
 (iv) Required probability $= P[\text{At least one head}]$
 $= 1 - P[\text{No head}] = 1 - p(0)$
 $= 1 - \frac{1}{1024} = \frac{1023}{1024}$ [From Part (iii)]
 (v) Required probability $= P(X \leq 3) = p(0) + p(1) + p(2) + p(3)$
 $= \frac{1}{1024} [{}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + {}^{10}C_3] = \frac{1 + 10 + 45 + 120}{1024} = \frac{176}{1024} = \frac{11}{64}$
 (vi) Required probability $= (X \geq 4) = p(4) + p(5) + \dots + p(10)$
 $= \frac{1}{1024} [{}^{10}C_4 + {}^{10}C_5 + \dots + {}^{10}C_{10}]$

Last part can be conveniently done as follows :

$$\text{Required probability} = P(X \geq 4) = 1 - P(X \leq 3)$$

$$= 1 - \left[p(0) + p(1) + p(2) + p(3) \right] = 1 - \frac{11}{64} = \frac{53}{64} \quad \text{[From Part (v)]}$$

Example 14.2. Define Binomial Distribution. What is the probability of guessing correctly at least six

Poisson Distribution

(Limiting Case of Binomial Distribution)

Poisson Distribution may be obtained as a limiting case of Binomial distribution under the following conditions:

- (i) n = no. of trials is indefinitely large, i.e., $n \rightarrow \infty$
- (ii) p = probability of success for each trial is indefinitely small, i.e. $p \rightarrow 0$
- (iii) $np = \lambda$ (say) is finite.

Then the probability function of the Poisson distribution is $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$, $x=0, 1, 2, \dots$

= Probability of x successes in n trials.

- Mean = $np = \lambda$
- Variance = $\sigma^2 = \mu_2 = np = \lambda$
- Mean = Variance = λ

Q:- It is known from past experience that in a certain plant there are on the average 4 industrial accidents per month. Find the probability that in a given year there will be less than 4 accidents. Assume Poisson distribution ($e^{-4} = 0.0183$)

Sol:- Here we are given $\lambda = 4$

Let X denote the no. of accidents in a plant per month.

Then probability that there will be less than 4 accidents is

$$\begin{aligned} P(X < 4) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= \frac{e^{-\lambda} \lambda^0}{0!} + \frac{e^{-\lambda} \lambda^1}{1!} + \frac{e^{-\lambda} \lambda^2}{2!} + \frac{e^{-\lambda} \lambda^3}{3!} \\ &= e^{-\lambda} \left[1 + \lambda + \frac{\lambda^2}{2} + \frac{\lambda^3}{6} \right] \\ &= e^{-4} \left[1 + 4 + 8 + \frac{32}{3} \right] \\ &= 0.0183 \times 23.67 = 0.4332 \end{aligned}$$

Q:- In a certain factory turning out razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10, use Poisson distribution to calculate the approximate no. of packets containing no defective, one defective and two defective blades respectively in a consignment of 10000 packets.

Sol:- Here $n = 10$,
 $P =$ Probability of defective blade $= 0.002$

$$\lambda = np = 10 \times 0.002 = 0.02$$

$$[e^{-\lambda} = e^{-0.02} = 0.9802]$$

No. of packets containing no defective blade

$$= 10000 \times P(X=0)$$

$$= 10000 \times \frac{e^{-\lambda} \lambda^0}{0!}$$

$$= 10000 \times 0.9802$$

$$= 9802.$$

No. of packets containing one defective blade

$$= 10000 \times P(X=1)$$

$$= 10000 \times \frac{e^{-\lambda} \lambda}{1!}$$

$$= 10000 \times e^{-0.02} \times 0.02$$

$$= 196$$

No. of packets containing two defective blades

$$= 10000 \times P(X=2)$$

$$= 10000 \times \frac{e^{-\lambda} \lambda^2}{2!}$$

$$= \frac{10000 \times e^{-0.02} (0.02)^2}{2}$$

$$= 2 \text{ (approx)}$$

Hence, the number of packets of containing 0, 1, 2, and 3 defective lenses is respectively 19604, 392, 4, 0.

Example 14-25. A car hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson variate with mean 1.5. Calculate the proportion of days on which

(i) Neither car is used

(ii) Some demand is refused.

Solution. Let the random variable X denote the number of demands for a car on any day. Then, X follows Poisson distribution with parameter $m = 1.5$.

$\therefore P(X = r)$ = Probability of r demands for a car on any day.

$$= \frac{e^{-1.5} (1.5)^r}{r!} \quad \dots (**)$$

(i) Neither car will be used, if there is no demand for any car. Hence the required proportion of days on which no car is used is given by : [From (**)]

$$\begin{aligned} P(X = 0) &= e^{-1.5} = \text{Antilog} [-1.5 \log_{10} e] \\ &= \text{Antilog} [-1.5 \log_{10} 2.718] = \text{Antilog} [-1.5 \times 0.4343] \\ &= \text{Antilog} [-0.65145] = \text{Antilog} [\bar{1} \cdot 34855] = 0.2231 \end{aligned}$$

(ii) Since the firm has only two cars, some demand will be refused if the number of demands per day is greater than 2. Hence, the required proportion is given by :

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - \left[e^{-1.5} + 1.5e^{-1.5} + \frac{(1.5)^2}{2!} e^{-1.5} \right] = 1 - e^{-1.5} \left[1 + 1.5 + \frac{2.25}{2} \right] \text{ [From (**)]} \\ &= 1 - 0.2231 (1 + 1.5 + 1.125) = 1 - 0.2231 \times 3.625 \\ &= 1 - 0.80874 = 0.19126. \end{aligned}$$

Example 14-26. Air Corporation having had just 2 air crashes during its first fifty years of service.